

GENAI -HANDSON-1

SRN: PES2UG23CS306

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SECTION: E

Task	Model	Classification	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
Generation	BERT	Failure	Generated nonsense or random symbols.	BERT is an Encoder. it isn't trained to predict the next word.
	RoBERTa	Failure	Did not generate any continuation beyond the prompt	Like BERT, RoBERTa is Encoder-only; and cannot generate new tokens sequentially
	BART	Failure	Generated output but text was noisy, repetitive, and semantically meaningless	It is an Encoder-Decoder (seq-to-seq) model designed for reconstruction and text generation.
Fill-Mask	BERT	Success	Correctly predicted context-appropriate words like "create", "generate", and	BERT is trained on Masked Language

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			"produce" with high confidence	Modeling (MLM).
	RoBERTa	Success	Predicted accurate words such as "generate" and "create" with strong confidence scores	An optimized version of BERT that uses dynamic masking for better MLM performance.
	BART	Success	Predicted reasonable but less precise words with much lower confidence	BART is primarily a sequence-to-sequence model and not optimized for MLM
QA	BERT	Success	Correctly identifies answer spans in text "hallucinations, bias, and deepfakes", but with a very low confidence score	Encoder-only model can understand context, but base BERT is not fine-tuned for QA tasks
	RoBERTa	Partial Success	Returned only a partial answer span ("such as hallucinations") with very low confidence	Strong encoder representations but lacks QA-specific fine-tuning

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	BART	Partial Success	Extracted only part of the correct answer("deepfakes") with low confidence	Encoder–decoder architecture requires QA fine-tuning to perform accurate span extraction